Lecture 30.05: AdaBoost Intuition — Decision Stumps & Sample Reweighting

Status Complete

Jupyter Notebook

Notes Hinglish

PDF English

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📌 Topic Overview

Comprehensive intuition for AdaBoost (Adaptive Boosting). Covers the school quiz teacher analogy, Email Spam classification dataset, uniform sample weights w_i = 1/N, decision stumps, stump voting power alpha, and sample reweighting.

📋 Key Learning Objectives

- What AdaBoost (Adaptive Boosting) is and its core intuition.
- Ma'am's School Teacher Quiz Analogy: How repeated mistakes receive higher attention.
- Real-World Binary Dataset: Email Spam Detection (email_length, number_of_links).
- Decision Stump: Why single-split trees (max_depth=1) are used as weak learners.
- Sample Weight Initialization: Starting with uniform weights w_i = 1/N.
- Stump Performance Score (alpha): Formula alpha = 0.5 * ln((1 - E) / E).
- Sample Reweighting Rule: Multiplying by exp(+alpha) for wrong samples, exp(-alpha) for right samples.
- Normalization Factor (Z_m): Ensuring weights sum back to 1.0.

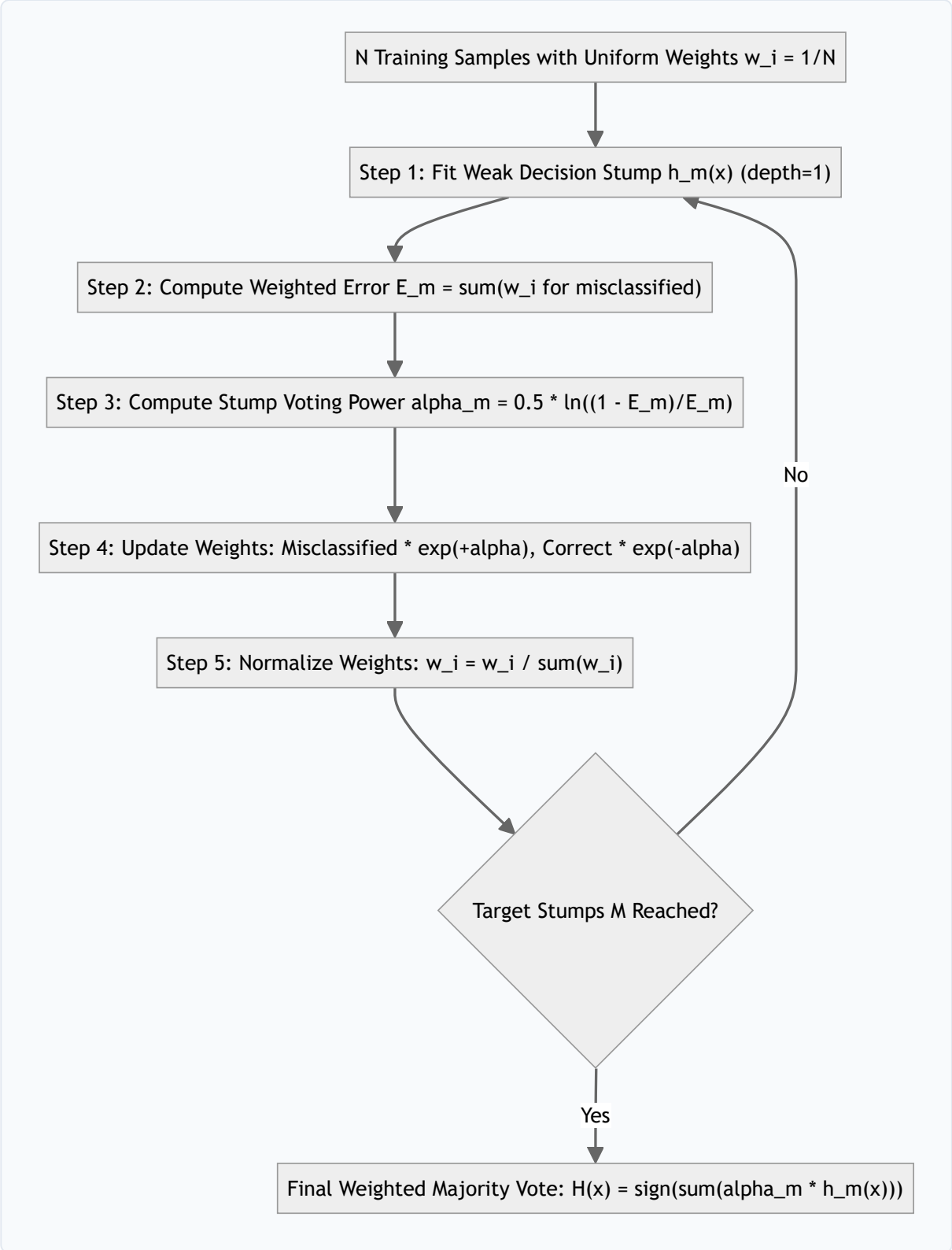
🔍 In-Depth Conceptual Breakdown

- **AdaBoost Ka Naam**: AdaBoost stands for **Adaptive Boosting** — yeh har stage par adapt hota hai pichhle models ki galtiyo ke mutabiq.

- **Ma'am Ki School Teacher Analogy**: Bachpan mein jab teacher chapter ke questions poochti thi, jo question student galat batata tha, teacher agle round mein usi question par sabse zyada focus karti thi! AdaBoost bhi bilkul wahi karta hai: jin samples ko model galat classify karta hai, unka weight badha deta hai.

- **Email Spam Detection Example**: Board par data explain kiya gaya: `Email Length`, `Number of Links`, aur target `Spam: Yes/No (1/0)`. Generally lambi emails aur zyada links wali emails spam hoti hain.
- **Decision Stump (Weak Learner)**: AdaBoost mein full trees nahi bante! Sirf 1 split wala chota tree banta hai jise **Decision Stump** kehte hain (`max_depth=1`).
- **Step 1: Uniform Initial Weights ($w_i = 1/N$)**: Maan lo dataset mein $N = 6$ emails hain. Har email ka starting weight hoga: $w_i = 1/6$. Total sum hamesha 1.0 rehta hai.
- **Step 2: Weighted Error (E)**: Stump 1 ne 6 mein se 1 email ko galat predict kiya. Total Error E = us galat sample ka weight = $1/6 \approx 0.167$.
- **Step 3: Stump Voting Power (α)**: Formula: $\alpha = \frac{1}{2} \ln\left(\frac{1-E}{E}\right)$. Agar $E = 0.167$, toh $\alpha \approx 0.805$. Kam error wale stump ka voting power zyada hota hai!
- **Step 4: Sample Reweighting Rule**:
  - **Correctly Classified Sample**: $w_{new} = w_{old} \times e^{-\alpha}$ (Weight kam ho jayega!).
  - **Misclassified Sample**: $w_{new} = w_{old} \times e^{+\alpha}$ (Weight exponentially badh jayega!).
- **Step 5: Normalize Weights (Z_m)**: Naye weights ka sum 1.0 nahi rehta, isliye unhe total sum $Z_m = \sum w_i$ se divide karke normalize karte hain: $w_{norm} = w_{new} / Z_m$.
- **Final Ensemble Prediction**: Saare stumps ka weighted vote: $H(x) = \text{sign}\left(\sum_{m=1}^{M} \alpha_m h_m(x)\right)$.

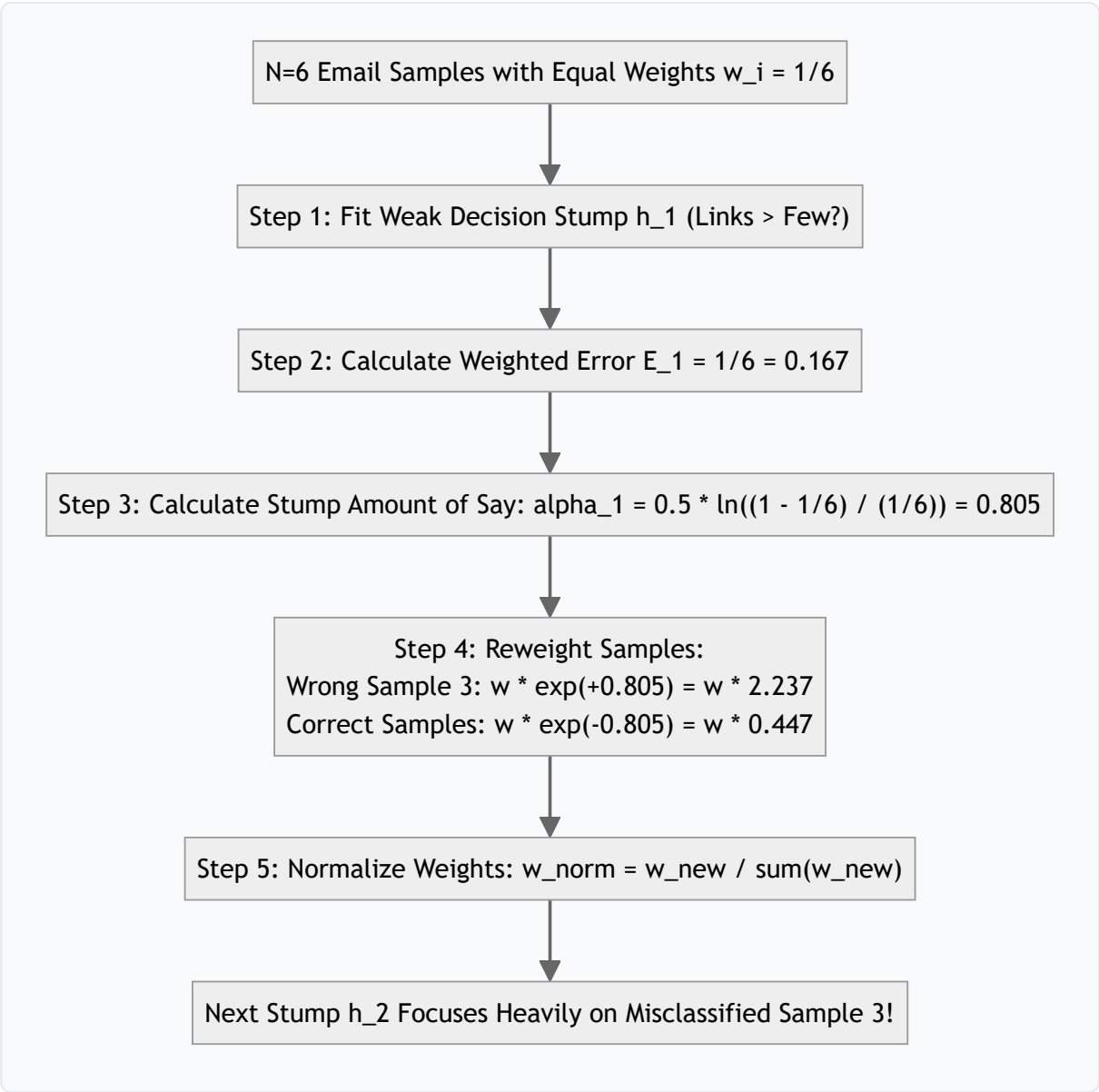
📊 Architectural Flowchart



📋 Ma'am's Classroom Blackboard Email Spam Reweighting Table

Email ID	Email Length	No. of Links	Actual Target: Spam? (y)	Initial Weight ($w_1 = 1/6$)	Stump 1 Split Prediction	Classification Status	Weight Update Direction
1	Long	Many	**+1 (Spam)**	1/6 (0.167)	+1 (Spam)	Correct	Multiplied by $e^{-\alpha}$ (Decreased)
2	Short	Few	**-1 (Not Spam)**	1/6 (0.167)	-1 (Not Spam)	Correct	Multiplied by $e^{-\alpha}$ (Decreased)
3	Long	Few	**-1 (Not Spam)**	1/6 (0.167)	+1 (Spam)	**WRONG (Misclassified!)**	**Multiplied by $e^{+\alpha}$ (Increased!)**
4	Short	Many	**+1 (Spam)**	1/6 (0.167)	+1 (Spam)	Correct	Multiplied by $e^{-\alpha}$ (Decreased)
5	Medium	Many	**+1 (Spam)**	1/6 (0.167)	+1 (Spam)	Correct	Multiplied by $e^{-\alpha}$ (Decreased)
6	Medium	Few	**-1 (Not Spam)**	1/6 (0.167)	-1 (Not Spam)	Correct	Multiplied by $e^{-\alpha}$ (Decreased)

📊 Ma'am's Decision Stump & Reweighting Architecture



🔢 Mathematical Formulation

Initial sample weight assignment for N observations:

$$w_i^{(1)} = \frac{1}{N}, \quad \forall i \in \{1, 2, \ldots, N\}$$

Weighted classification error rate E_m of stump h_m:

$$E_m = \sum_{i=1}^{N} w_i^{(m)} \cdot \mathbb{I}\left(y_i \neq h_m(x_i)\right)$$

Stump voting weight (amount of say) α_m:

$$\alpha_m = \frac{1}{2} \ln \left(\frac{1 - E_m}{E_m} \right)$$

Weight update rule with normalization constant Z_m:

$$w_i^{(m+1)} = \frac{w_i^{(m)} \exp\left(-\alpha_m y_i h_m(x_i)\right)}{Z_m}, \quad \text{where } Z_m = \sum_{i=1}^{N} w_i^{(m+1)}$$

💻 Implementation

The fully tested and verified standalone implementation is provided in the synchronized Jupyter Notebook.